

Cutoff for random Cayley graphs of finite groups

Abstract. We show that the continuous-time random walk on a Cayley graph generated by independent uniform elements exhibits cutoff with high probability whenever the number of generators grows faster than the logarithm of the group order and is subpolynomial in that order. This resolves a longstanding conjecture of Aldous and Diaconis. We give a deterministic cutoff time in terms of the expected entropy of the walk; this time may depend on the group structure.

1. Introduction

The cutoff phenomenon describes a sharp transition in the convergence of a Markov chain to stationarity: the total variation distance falls from near one to near zero in a time interval of smaller order than the mixing time. Classical examples arise from card shuffling and random walks on finite groups¹. Aldous and Diaconis² asked whether this phenomenon is typical for Cayley graphs with random generators. The conjecture predicts cutoff with high probability for a group of order n with k independent uniform generators whenever $k/\log n \rightarrow \infty$ and $\log k/\log n \rightarrow 0$; see the formulation of Huang and Rao³.

Early cutoff results for abelian groups were obtained by Hildebrand⁴. Dou and Hildebrand⁵ gave general upper bounds and matching lower bounds for several classes of groups. Further

¹ P. Diaconis and M. Shahshahani. "Generating a random permutation with random transpositions". In: *Z. Wahrsch. Verw. Gebiete* 57.2 (1981), pp. 159–179; D. Aldous and P. Diaconis. "Shuffling cards and stopping times". In: *Amer. Math. Monthly* 93.5 (1986), pp. 333–348.

² D. Aldous and P. Diaconis. *Shuffling cards and stopping times*. Technical Report 231. Department of Statistics, Stanford University, 1985, p. 40.

³ X. Huang and R. Rao. *Cutoff for random walks on dihedral groups*. 2025. arXiv: 2510.19942, Conjecture 1.

⁴ M. Hildebrand. "Random walks supported on random points of $\mathbb{Z}/n\mathbb{Z}$ ". In: *Probab. Theory Related Fields* 100.2 (1994), pp. 191–203, Theorem 3 and p. 202.

⁵ C. Dou and M. Hildebrand. "Enumeration and random random walks on finite groups". In: *Ann. Probab.* 24.2 (1996), pp. 987–1000.

bounds were obtained by Roichman⁶; see Hildebrand⁷ for a survey. For arbitrary finite groups, cutoff was already known when $\log \log n \ll \log k \ll \log n$.⁸ Hermon and Olesker-Taylor⁹ extended the abelian results below logarithmic degree, under an almost necessary condition on the number of generators. For non-abelian groups, the cutoff time need not depend only on the group order and the number of generators, as their work on upper triangular groups shows¹⁰.

Hermon and Huang¹¹ proved cutoff for finite nilpotent groups of bounded rank and nilpotency class in the range $1 \ll \log k \ll \log n$. Huang and Rao¹² treated dihedral groups of twice prime order in nearly this range and, when also $k/\log n \rightarrow \infty$, groups with abelian normal subgroups of uniformly bounded index. We prove cutoff for arbitrary finite groups in the conjectured range. Our cutoff time is deterministic and may depend on the group structure.

1.1 Random Cayley graphs

Let G be a finite group of order n with identity e , and write $[k] = \{1, \dots, k\}$. Let $Z_1, \dots, Z_k$ be independent uniform elements of G , and write $Z = (Z_1, \dots, Z_k)$. For a fixed tuple $z = (z_1, \dots, z_k)$, we consider the Cayley multigraph on G whose step multiset consists of the $2k$ elements z_i^ε , where $i \in [k]$ and $\varepsilon \in \{-1, 1\}$. We retain all multiplicities, including those arising from repeated generators, inverse pairs, involutions, and loops.¹³

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- ⁶ Y. Roichman. "On random random walks". In: *Ann. Probab.* 24.2 (1996), pp. 1001–1011.
- ⁷ M. Hildebrand. "A survey of results on random random walks on finite groups". In: *Probab. Surv.* 2 (2005), pp. 33–63.
- ⁸ J. Hermon and S. Olesker-Taylor. "Cutoff for almost all random walks on abelian groups". In: *J. Eur. Math. Soc.* 28 (2026), pp. 1913–1977, Section 1.3.1 and Theorem 7.2.
- ⁹ J. Hermon and S. Olesker-Taylor. "Cutoff for almost all random walks on abelian groups". In: *J. Eur. Math. Soc.* 28 (2026), pp. 1913–1977, Theorem A.
- ¹⁰ J. Hermon and S. Olesker-Taylor. *Cutoff for random walks on upper triangular matrices*. Revised 2021. 2019. arXiv: 1911.02974, Theorem A.
- ¹¹ J. Hermon and X. Huang. *Cutoff for random Cayley graphs of nilpotent groups*. 2024. arXiv: 2403.12355, Theorem 1.
- ¹² X. Huang and R. Rao. *Cutoff for random walks on dihedral groups*. 2025. arXiv: 2510.19942, Theorems 1.3 and 1.4.
- ¹³ X. Huang and R. Rao. *Cutoff for random walks on dihedral groups*. 2025. arXiv: 2510.19942, Remark 1.2.

The walk starts at $\mathbf{e}$ and, at rate one, multiplies its current position on the right by a uniformly chosen element of this multiset. The corresponding discrete-time transition matrix is

$$P_z(x, y) = \frac{1}{2k} \sum_{i=1}^k \sum_{\varepsilon \in \{-1, 1\}} 1_{\{y=xz_i^\varepsilon\}}. \quad (1.1)$$

Write π for the uniform probability measure on G and $q_{z,m} = P_z^m(\mathbf{e}, \cdot)$ for the distribution after m steps. The continuous-time distribution and its distance from stationarity are

$$p_{z,t} = e^{-t} \sum_{m=0}^{\infty} \frac{t^m}{m!} q_{z,m}, \quad d_z(t) = \|p_{z,t} - \pi\|_{TV}. \quad (1.2)$$

Here $\|\nu\|_{TV} = \frac{1}{2} \sum_x |\nu(x)|$ for a signed measure ν . Since left multiplication preserves the transition probabilities, $d_z(t)$ is independent of the starting vertex. For $\varepsilon \in (0, 1)$, let $t_{\min, z}(\varepsilon) = \inf\{t \geq 0 : d_z(t) \leq \varepsilon\}$, with value infinity if the set is empty. Unless otherwise stated, convergence in probability refers to the choice of Z .

Let $\mathfrak{A} = [k] \times \{-1, 1\}$ be the alphabet of signed indices. For each non-negative integer m , write $\mathcal{W}_m = \mathfrak{A}^m$ for the set of words of length m . A word $w = ((i_1, \varepsilon_1), \dots, (i_m, \varepsilon_m))$ determines a walk from $\mathbf{e}$ with endpoint $w(z) = z_{i_1}^{\varepsilon_1} \dots z_{i_m}^{\varepsilon_m}$. The empty word has value $\mathbf{e}$. A uniform word has independent letters, each uniform on $\mathfrak{A}$. Throughout, C denotes a positive absolute constant that may change from one occurrence to the next.

1.2 Main results

For a probability measure q on a finite set, write $H(q) = -\sum_x q(x) \log q(x)$, with $0 \log 0 = 0$. All logarithms are natural. We define the expected entropy after m steps by

$$h_m = \mathbb{E}_Z H(q_{Z,m}), \quad m \geq 0. \quad (1.3)$$

Here we average the entropies for fixed tuples of generators. Define

$$t_*(G, k) = \min\{m \in \mathbb{Z}_{\geq 0} : h_m \geq \log n - 2(\log n)^{2/3}\}, \quad (1.4)$$

with value infinity if the set is empty. The following theorem shows that this entropy threshold gives a cutoff time.

Theorem 1.1. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy

$$\frac{k}{\log n} \rightarrow \infty, \quad \frac{\log k}{\log n} \rightarrow 0. \quad (1.5)$$

For each G , let Z consist of k independent uniform elements of G . Then $t_*(G, k)$ is finite for all sufficiently large n and tends to infinity. Moreover, for the walk defined by (1.1)–(1.2) and every fixed $\delta \in (0, 1)$,

$$d_Z((1 - \delta)t_*) \xrightarrow{\mathbb{P}} 1, \quad d_Z((1 + \delta)t_*) \xrightarrow{\mathbb{P}} 0. \quad (1.6)$$

Combining Theorem 1.1 with the monotonicity of d_Z , we obtain the following equivalent statement in terms of mixing times.

Corollary 1.2. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy $k/\log n \rightarrow \infty$ and $\log k/\log n \rightarrow 0$. For each G , let Z consist of k independent uniform elements of G . Then, for every fixed $\varepsilon \in (0, 1)$,

$$\frac{t_{\text{mix}, Z}(\varepsilon)}{t_*(G, k)} \xrightarrow{\mathbb{P}} 1. \quad (1.7)$$

The main step in the proof is to bound the variance of the information content of the walk. For a uniform word W in $\mathcal{W}_m$, independent of Z , let

$$I_m = -\log q_{Z, m}(W(Z)). \quad (1.8)$$

Our next theorem gives a bound that is uniform over all finite groups.

Theorem 1.3. Let G be a finite group, and let k, m be positive integers with $2 \leq m \leq k/4096$. Let Z consist of k independent uniform elements of G , and let W be an independent uniform word in $\mathcal{W}_m$. Then

$$\text{Var}_{Z, W}(I_m) \leq C m \log^6(2k), \quad (1.9)$$

where C is an absolute constant.

Since $\mathbb{E}[I_m | Z] = H(q_{Z, m})$, the conditional variance formula gives

$$\text{Var}_{Z, W}(I_m) = \mathbb{E}_Z \text{Var}_{X \sim q_{Z, m}}(-\log q_{Z, m}(X)) + \text{Var}_Z(H(q_{Z, m})). \quad (1.10)$$

Theorem 1.3 thus governs both the information fluctuations for a fixed choice of generators, averaged over that choice, and the variation of entropy between choices of generators.

Hermon, Huang, Pedrotti, and Salez¹⁴ proved a variance bound for a fixed symmetric, conjugacy-invariant step distribution.

1.3 Proof strategy

Our main step is the variance estimate in Theorem 1.3. We adapt the Efron–Stein argument used by Eberhard and Varjú¹⁵ and Huang and Rao¹⁶ to bound the variance under the joint law of the generators and the walk.

We construct independent coordinates for the generators and the word. At a position whose index occurs only once in the word, we prescribe the product up to that position as an independent uniform element of $\mathbf{G}$. The preceding partial product and the prescribed value then determine the generator at that position. Together with independent uniform values for the remaining generators, this gives a uniform tuple in $\mathbf{G}^k$ for each fixed word.

In these variables, resampling one coordinate affects only a small collection of contiguous subwords, with high probability. The products on these subwords are preserved, except possibly on a final suffix, where the change amounts to a common right translation. We record the subwords and all occurrences of their affected indices. With the two tuples of group elements fixed, the product identity then holds for every word pair with that record. Hence it gives an identity between the conditional endpoint distributions. The mixture estimates in Lemmas 2.4–2.6 turn this identity into a bound on the change in information. We then apply the Efron–Stein inequality to obtain Theorem 1.3.

For $\log k > (\log \log n)^2$, a bound on the size of a ball in the Cayley graph and a collision estimate already give matching bounds on the mixing time. It therefore remains to treat $\log k \leq (\log \log n)^2$. In this range, we transfer Theorem 1.3 to continuous time and combine it with the spectral estimate to obtain a cutoff window of length $O((\log n)^{2/3})$ around t_* .

The paper is organised as follows. In Section 2, we construct independent coordinates from partial products and prove the conditional distribution identity needed for resampling. In Section 3, we prove Theorem 1.3. In Section 4, we prove

¹⁴ J. Hermon, X. Huang, F. Pedrotti, and J. Salez. “Concentration of information on discrete groups”. In: *Probab. Theory Related Fields* (2026). Published online. DOI: 10.1007/s00440-026-01498-6.

¹⁵ S. Eberhard and P. P. Varjú. “Mixing time of the Chung–Diaconis–Graham random process”. In: *Probab. Theory Related Fields* 179.1–2 (2021), pp. 317–344. DOI: 10.1007/s00440-020-01009-1, Section 2.

¹⁶ X. Huang and R. Rao. *Cutoff for random walks on dihedral groups*. 2025. arXiv: 2510.19942, Section 3.4.2.

Theorem 1.1 using the information bound and estimates on mixing times.

Use of generative AI. During the development of this work, the author used generative AI to explore an approach to cutoff based on collision probabilities. The model suggested the initial structure of the argument, leaving the required concentration result unproved. The author subsequently reconstructed and extended the argument to cover the full parameter range considered in this paper, independently verified the resulting proof, and prepared the manuscript.

2. Words and conditional distributions

We construct the independent coordinates used in Theorem 1.3 and compare conditional endpoint distributions after resampling. The comparison uses products on fixed intervals and complete occurrence lists of the affected indices. Indices are distinguished even when their group values coincide.

2.1 Partial products

A partial product at position r is the product of the first r letters, evaluated at a given tuple of generators. For an index occurring only once, we may prescribe this product at its position and solve for its generator. We use these prescribed products as independent coordinates.

For $w = ((i_1, \varepsilon_1), \dots, (i_m, \varepsilon_m))$, let

$$N_i(w) = \#\{r : i_r = i\}, \quad J_i(w) = \{(r, \varepsilon_r) : i_r = i\}.$$

The list $J_i(w)$ contains every occurrence of i , with its sign, in increasing order of position. An index is a *singleton index* if it occurs exactly once in w . A position r is a *singleton position* if $N_{i_r}(w) = 1$. Write $S(w)$ for the set of singleton positions. If $S(w) = \{j_1 < \dots < j_s\}$, write

$$w = c_0 a_1 c_1 a_2 \cdots a_s c_s, \tag{2.1}$$

where $a_\ell = w_{j_\ell}$ and each c_ℓ is a possibly empty word. Every index appearing in a c_ℓ occurs at least twice in w .

Let $B_1, \dots, B_k, U_1, \dots, U_m$ be independent uniform elements of G , independent also of the uniform word W , and write $B = (B_1, \dots, B_k)$ and $U = (U_1, \dots, U_m)$. For each fixed word w , we construct $\Psi_w(B, U)$ by keeping $Z_i = B_i$ for every index i that is not singleton, and prescribing the partial products at singleton positions. More precisely, at position j_ℓ let

$$Z_{i_{j_\ell}}^{\varepsilon_{j_\ell}} = c_{\ell-1}(B)^{-1} U_{j_{\ell-1}}^{-1} U_{j_\ell}, \quad U_{j_0} = e. \tag{2.2}$$

Here $j_0 = 0$ is the initial boundary. This definition gives $\Psi_w(B, U) = B$ when there are no singleton positions. The coordinates U_j with $j \notin S(w)$ and the coordinates B_i for singleton indices are not used.

Lemma 2.1. Let w be a word of length m . Then $\Psi_w(B, U)$ is uniform on G^k , and the first j_ℓ letters of w , evaluated at this tuple, have product U_{j_ℓ} . Consequently, $Z = \Psi_W(B, U)$ is independent of W .

Proof. By (2.2), $c_{\ell-1}(B)Z_{i_{j_\ell}}^{\varepsilon_{j_\ell}} = U_{j_{\ell-1}}^{-1}U_{j_\ell}$. Multiplication in the order of (2.1) shows that the partial product at j_ℓ is U_{j_ℓ} .

Now fix the coordinates B_i for the non-singleton indices. The generator values at singleton indices are determined successively by $(U_{j_1}, \dots, U_{j_s})$, and the partial products recover these coordinates from the generator values. Taking an inverse for a negative sign is also a bijection. These generator values are therefore conditionally independent and uniform. Together with the values at non-singleton indices, they form a uniform tuple on G^k . This law is the same for every w , so the resulting tuple is independent of W . $\square$

Remark 2.2. The argument also applies after resampling one coordinate. After resampling B_i or U_j , each of the two tuples is uniform conditional on W . After resampling W_j , conditional on (W, W') , each of $\Psi_W(B, U)$ and $\Psi_{W'}(B, U)$ is uniform. Thus each tuple separately is independent of the word pair.

The number and length of the contiguous subwords affected by a resampling will be bounded in terms of multiplicities and gaps between singleton positions. For $m \geq 2$, let $b = \lceil 100 \log(m+1) \rceil$. Call a word of length m *good* if every index has multiplicity at most b and every interval containing no singleton position has length at most b . Write $\mathcal{G}_{k,m}$ for the set of good words.

Lemma 2.3. Let $2 \leq m \leq k/4096$, and let W be uniform on $\mathcal{W}_m$. Then

$$\mathbb{P}(W \notin \mathcal{G}_{k,m}) \leq Cm^{-40}, \quad (2.3)$$

where C is an absolute constant.

Proof. Every word is good when $b \geq m$, so suppose that $b < m$. We first bound the probability of a large multiplicity. Choosing one position and then $b-1$ further positions with the same index gives the bound

$$m \binom{m-1}{b-1} k^{-(b-1)} \leq m \left(\frac{em}{(b-1)k} \right)^{b-1}$$

for the probability that an index has multiplicity at least $\mathbf{b}$. This also bounds the probability of failure of the multiplicity condition.

For the gap condition, fix an interval of $\mathbf{b}$ positions and expose all indices outside it. There are at most $\mathbf{m}$ such indices. If the interval contains no singleton, either at least $\mathbf{b}/2$ entries use these indices, or more than $\mathbf{b}/2$ use indices absent outside the interval. In the latter case, each new index must occur at least twice inside the interval, so at least $\mathbf{b}/4$ positions repeat an index already seen there.

The first alternative has probability at most $2^{\mathbf{b}}(\mathbf{m}/\mathbf{k})^{\mathbf{b}/2}$. For the second, expose the interval from left to right. At each specified position the conditional probability of a repetition is at most $\mathbf{b}/\mathbf{k}$, so a union bound gives $2^{\mathbf{b}}(\mathbf{b}/\mathbf{k})^{\mathbf{b}/4}$. Integer rounding only improves these estimates. Since $\mathbf{m}/\mathbf{k} \leq 1/4096$ and $\mathbf{b} \leq \mathbf{m}$, their sum is at most $\mathbf{C}4^{-\mathbf{b}}$. Taking a union bound over the at most $\mathbf{m}$ intervals, and combining this with the multiplicity estimate, proves (2.3) for the chosen value of $\mathbf{b}$. $\square$

2.2 Conditional distributions

Conditioning replaces an endpoint distribution by one component of a mixture. The following lemma controls the resulting logarithmic ratio. For the first estimate, we follow the argument of Eberhard and Varjú¹⁷ and Huang and Rao¹⁸.

Lemma 2.4. Let $\mathbf{p}, \mathbf{q}$ be probability measures on a finite set, and let $0 < \alpha \leq 1$. Suppose that $\mathbf{p}(\mathbf{x}) \geq \alpha \mathbf{q}(\mathbf{x})$ for every $\mathbf{x}$. Then, for $\mathbf{X} \sim \mathbf{q}$,

$$\mathbb{E} \left[\log^2 \frac{\mathbf{p}(\mathbf{X})}{\mathbf{q}(\mathbf{X})} \right] \leq \log^2(1/\alpha) + 2. \quad (2.4)$$

If $\mathbf{p} = \alpha \mathbf{q} + (1 - \alpha) \mathbf{r}$ for a probability measure $\mathbf{r}$ and $\alpha \geq 1/2$, then

$$\mathbb{E} \left[\log^2 \frac{\mathbf{p}(\mathbf{X})}{\mathbf{q}(\mathbf{X})} \right] \leq \mathbf{C}(1 - \alpha), \quad (2.5)$$

where $\mathbf{C}$ is an absolute constant.

Proof. The logarithm is at least $\log \alpha$. Since $\mathbb{E}_{\mathbf{q}}[\mathbf{p}(\mathbf{X})/\mathbf{q}(\mathbf{X})] = \sum_{\mathbf{q}(\mathbf{x}) > 0} \mathbf{p}(\mathbf{x}) \leq 1$, its positive tail at $\mathbf{s} \geq 0$ is at most

¹⁷ S. Eberhard and P. P. Varjú. "Mixing time of the Chung-Diaconis-Graham random process". In: *Probab. Theory Related Fields* 179.1-2 (2021), pp. 317-344. DOI: 10.1007/s00440-020-01009-1, Lemma 2.2.

¹⁸ X. Huang and R. Rao. *Cutoff for random walks on dihedral groups*. 2025. arXiv: 2510.19942, proof of Proposition 3.14.

e^{-s} . The negative part therefore contributes at most $\log^2(1/\alpha)$ to the second moment, and the positive part contributes at most $\int_0^\infty 2se^{-s} ds = 2$.

For the second assertion, $-\log \alpha \leq 2(1-\alpha)$. The inequality $\log^2 x \leq C(x-1)$ for $x \geq 1$ bounds the positive contribution by

$$C \sum_x (p(x) - q(x))_+ = C(1-\alpha) \sum_x (r(x) - q(x))_+ \leq C(1-\alpha).$$

The negative contribution is at most $4(1-\alpha)^2$, giving (2.5). $\square$

The following bound lets us sum this estimate over the values of a finite-valued random variable.

Lemma 2.5. Let A be a finite-valued random variable, and let $\mathcal{D}$ be a set of at most M values of A with total probability $p > 0$. Then

$$\sum_{a \in \mathcal{D}} \mathbb{P}(A = a) \log^2 \frac{1}{\mathbb{P}(A = a)} \leq Cp \left(1 + \log^2 \frac{M}{p} \right), \quad (2.6)$$

where values of probability zero are omitted.

Proof. For $a \in \mathcal{D}$, let $\beta_a = \mathbb{P}(A = a)/p$. Under the distribution (β_a) , the event $\log(1/\beta_a) > \log M + s$ has probability at most e^{-s} for $s \geq 0$. Integration gives $\mathbb{E}_\beta \log^2(1/\beta_a) \leq C(1 + \log^2 M)$. The result follows from $\log(1/\mathbb{P}(A = a)) = \log(1/p) + \log(1/\beta_a)$ and $(x+y)^2 \leq 2x^2 + 2y^2$. $\square$

The next lemma compares information when a bijection of G maps one conditional endpoint distribution to the other. As in Remark 2.2, each tuple of generators is required to be independent of the word pair.

Lemma 2.6. Let (Z, W, Z', W') be coupled random variables such that W, W' are uniform on $\mathcal{W}_m$, and each of Z, Z' is uniform on G^k and independent of (W, W') . Let A be a function of (W, W') , and write

$$Q_Z^a(x) = \mathbb{P}(W(Z) = x \mid A = a), \quad Q_{Z'}^a(x) = \mathbb{P}(W'(Z') = x \mid A = a)$$

whenever $\mathbb{P}(A = a) > 0$. Suppose that, on $\{A = a\}$, there is a bijection $\phi: G \rightarrow G$ such that

$$W'(Z') = \phi(W(Z)), \quad Q_{Z'}^a(\phi(x)) = Q_Z^a(x) \quad \text{for all } x \in G. \quad (2.7)$$

Let $F = -\log q_{Z,m}(W(Z))$ and $F' = -\log q_{Z',m}(W'(Z'))$. Then

$$\mathbb{E}[(F - F')^2 \mid A = a] \leq 4 \log^2 \frac{1}{\mathbb{P}(A = a)} + 8. \quad (2.8)$$

If (2.7) holds for at most M specified values of A with total probability $p > 0$, their contribution to $\mathbb{E}(F - F')^2$ is at most $Cp(1 + \log^2(M/p))$.

Proof. Let $\alpha = \mathbb{P}(A = \mathbf{a})$. Conditioning on A gives $q_{z,m} \geq \alpha Q_z^{\mathbf{a}}$ and $q_{z,m} \geq \alpha Q_z^{\mathbf{a}}$ for every fixed z . By independence from (W, W') , the endpoint law given the corresponding tuple and $A = \mathbf{a}$ is the conditional measure in the statement. We may therefore apply Lemma 2.4 to the two logarithms

$$\log \frac{q_{z,m}(W(Z))}{Q_z^{\mathbf{a}}(W(Z))} \quad \text{and} \quad \log \frac{q_{z',m}(W'(Z'))}{Q_{z'}^{\mathbf{a}}(W'(Z'))}$$

and bound each second moment by $\log^2(1/\alpha) + 2$. By (2.7), the denominators agree, so the difference of the two logarithms is $F' - F$. The inequality $(x - y)^2 \leq 2x^2 + 2y^2$ gives (2.8). Multiplying by the probabilities of the specified values of A , summing and applying Lemma 2.5 proves the final assertion. $\square$

2.3 Products on fixed intervals

For $0 \leq \mathbf{a} < \mathbf{c} \leq \mathbf{m}$, write $w_{(\mathbf{a}, \mathbf{c}]}$ for the restriction of w to positions $\mathbf{a} + 1, \dots, \mathbf{c}$, including both indices and signs. We shall apply Lemma 2.6 using a single bijection for every word pair allowed by the conditioning. The next lemma constructs this bijection as a right translation.

Lemma 2.7. Let $\mathcal{F}$ be a collection of pairs of words of length $\mathbf{m}$, and let $I_1, \dots, I_r$ be fixed disjoint intervals of positions, ordered from left to right. Suppose that each pair $(v, v') \in \mathcal{F}$ agrees outside these intervals, and that the two restrictions to each I_ℓ are fixed throughout $\mathcal{F}$. Let $D \subseteq [k]$, and suppose that every occurrence of an index in D , in either word of every pair, lies in $\bigcup_{\ell=1}^r I_\ell$.

Let $z, z' \in G^k$ agree outside D . Suppose that the products of the two restrictions, evaluated respectively at z, z' , agree on each I_ℓ , except possibly on I_r if it ends at $\mathbf{m}$. Then there is a right translation ϕ of G such that

$$v'(z') = \phi(v(z)) \quad \text{for every } (v, v') \in \mathcal{F}. \quad (2.9)$$

For any probability distribution on $\mathcal{F}$, the same translation maps one endpoint distribution to the other.

Proof. Outside the specified intervals, corresponding products agree: the words have the same letters there, and no index in D occurs. The interval products agree by assumption, with the possible exception of the final suffix. Thus the identity map suffices unless that suffix is exceptional. In that case its two products, say $\mathbf{a}, \mathbf{a}'$, are fixed throughout $\mathcal{F}$. Every endpoint pair has the form $x\mathbf{a}, x\mathbf{a}'$, so $\phi(y) = y\mathbf{a}^{-1}\mathbf{a}'$ works for every pair. Taking distributions gives the final assertion. $\square$

2.4 Resampling the coordinates

We now choose the intervals needed for each type of coordinate. A *record* will specify their positions and letters, complete occurrence lists, and the conditions defining the case under consideration. It contains no group values. Its fibre is the set of all word pairs with the specified data. Requiring a word to be good restricts this fibre without fixing the whole word.

Proposition 2.8. Let $2 \leq m \leq k/4096$, and use the construction $\Psi_W(B, U)$. Consider replacing one of the following coordinates by an independent copy:

1. U_j , on the event that W is good and $j \in S(W)$;
2. B_i , on the event that W is good and $N_i(W) \geq 2$;
3. a letter W_j , on the event that both W and the resulting word W' are good.

For each fixed coordinate, there is a record A , depending only on (W, W') , with at most

$$(2km)^{Cb^2} \tag{2.10}$$

values on the stated event, for which (2.7) holds on every corresponding fibre of positive probability. Outside that event, A takes a separate value $\dagger$. In the first two cases $W' = W$. In all three cases, each tuple is independent of (W, W') and hence of A .

Proof. For each resampling, fix a representative word pair and the coupled group coordinates. We keep the two resulting tuples fixed and choose a record for which Lemma 2.7 applies to every pair in its fibre. Include the stated good event in the record, and put $A = \dagger$ outside that event.

Resampling U_j . Suppose first that j is not the last singleton position, and let r be the next one. Resampling U_j changes only the generators at these two positions. Let g, h be the values of the letters at these positions, and let C_0 be the product of the intervening letters. The defining identities give

$$gC_0h = g'C_0h'. \tag{2.11}$$

The first factor is a fixed left multiple of U_j , the intervening generators are unchanged, and the last factor is $C_0^{-1}U_j^{-1}U_r$, so the occurrences of U_j cancel.

Record $[j, r] = (j-1, r]$, its restriction, the fact that r is the next singleton position, and the complete occurrence lists of the two endpoint indices. Each list has one entry. All occurrences of the

changed generators therefore lie in this interval for every word in the fibre. Its restriction is fixed, so (2.11) holds in the fixed tuples throughout the fibre. Lemma 2.7 gives equality of the endpoints.

If j is the last singleton position, record this fact, the suffix $[j, m]$, its restriction and the occurrence list of the index at j . Only the suffix product can change. The same lemma gives a common right translation. Since the word is good, the recorded interval has at most $b+2$ positions in the first case and $b+1$ in the second.

Resampling B_i . Suppose that $N_i(w) \geq 2$. Cut w immediately after each singleton position, into the intervals $(0, j_1], (j_1, j_2], \dots, (j_{s-1}, j_s]$ and the suffix $(j_s, m]$ when nonempty, and mark those containing i . There are at most b marked intervals, each of length at most $b+1$. Resampling B_i changes only Z_i and the generators at those right endpoints of marked intervals that are singleton positions. For an interval ending at such a position, (2.2) gives

$$c_\ell(B)Z_{i_{j_{\ell+1}}}^{\varepsilon_{j_{\ell+1}}} = u_{j_\ell}^{-1}u_{j_{\ell+1}}. \quad (2.12)$$

The partial products at the endpoints are unchanged, so this product is preserved. For the initial interval, the partial product at position zero is e . Only the final suffix may have a different product.

Record the marked intervals and their restrictions, identify the singleton right endpoints, and include $J_i(w)$ and the complete occurrence lists of these singleton indices. These lists confine every occurrence of a changed generator to a marked interval. The restrictions are fixed, so their products in the fixed tuples satisfy (2.12) for every word in the fibre. The final suffix is the only possible exception. This verifies all the hypotheses of Lemma 2.7.

Resampling W_j . Let w' be obtained from w by replacing the letter at position j , with B, U unchanged. Write a, c for the old and new indices. Only the multiplicities of a, c can change. Apart from j , a singleton position can be lost or gained only at the other occurrence of a when $N_a(w) = 2$, or at the unique occurrence of c when $N_c(w) = 1$. Thus

$$|S(w) \triangle S(w')| \leq 3. \quad (2.13)$$

If $a = c$, the sets of singleton positions are unchanged.

Use $(S(w) \cap S(w')) \setminus \{j\}$ as the common cuts, retaining the initial interval and any final suffix. Mark the intervals containing j or a position in $S(w) \triangle S(w')$. There are at most four. At most four

cuts after singleton positions of w have been removed; each new interval therefore joins at most five old intervals. Since w is good, every marked interval has length at most $5(b+1)$.

The partial products at the endpoints of a marked interval $(r, s]$ ending at a common singleton position are the same in both words. Hence

$$w_{(r,s]}(Z) = U_r^{-1} U_s = w'_{(r,s]}(Z'), \quad (2.14)$$

where $U_0 = e$. To apply Lemma 2.7, we must also locate every occurrence of a changed generator. An unmarked interval contains no changed letter or changed singleton position. If it ends at a singleton s , its left endpoint is the immediately preceding singleton in both words, or the initial boundary. The intervening word and the two boundary coordinates are the same, so (2.2) leaves the generator at its right endpoint unchanged.

For an index among a, c that is non-singleton in both words, the generator value is unchanged, including at any occurrences outside the marked intervals. Thus a generator value can change only at an index in the following set D : those among a, c that are singleton in at least one word, and the indices at common singleton positions that are right endpoints of marked intervals. All their occurrences lie in marked intervals. For a, c , this follows from the description of the changed singleton positions; a common right endpoint index occurs just once in each word.

Record j , the old and new letters, the marked intervals with their two restrictions, and which right endpoints are common singleton positions. Include the complete occurrence lists of a, c and of these right endpoint indices in both words, together with the condition that both words are good.

The record determines D . In every pair in its fibre, no occurrence of an index in D lies outside the marked intervals. The two fixed tuples agree outside D , and the fixed restrictions to the marked intervals satisfy (2.14) in those tuples. Since corresponding words also agree outside the marked intervals, Lemma 2.7 gives a single right translation for every pair in the fibre.

It remains to bound the number of records and deduce the distributional assertion. List the entries of a record in a fixed order. Each entry specifies a position, an index, a sign or one of finitely many cases, and so has at most Ckm possible values. The interval restrictions use $O(b^2)$ entries when resampling B_i and $O(b)$ in the other cases. Complete occurrence lists add $O(b)$ entries: at most b singleton indices are needed when resampling B_i ; when resampling W_j , there are at most six relevant indices, with only

$\mathbf{a}, \mathbf{c}$ possibly repeated. Including the number of entries and the stated conditions, at most $\mathbf{C}b^2$ entries suffice. This gives (2.10) after increasing $\mathbf{C}$.

For each record value other than $\dagger$, apply the identity on its whole fibre to the conditional law of (W, W') . The two marginals are the word laws defining $Q_z^{\mathbf{a}}$ and $Q_z'^{\mathbf{a}}$, so both identities in (2.7) follow. Each tuple is independent of (W, W') by Remark 2.2, and hence independent of its record. This proves the proposition. $\square$

Remark 2.9. The occurrence lists are needed to exclude changed generators outside the recorded intervals. Without this condition, an identity for the sampled word pair need not hold throughout the record fibre, so the conditional endpoint distributions need not correspond.

3. Concentration of information

We now prove the information bound using the Efron-Stein inequality.¹⁹ If F is a square-integrable function of independent random variables $Y_1, \dots, Y_d$, and $F^{(j)}$ is obtained by replacing Y_j with an independent copy, then

$$\text{Var } F \leq \frac{1}{2} \sum_{j=1}^d \mathbb{E}(F - F^{(j)})^2. \quad (3.1)$$

Thus it suffices to bound the sum of the mean-square changes under resampling. There are $\mathbf{m}$ coordinates U_j , $\mathbf{m}$ letters W_j and $\mathbf{k}$ coordinates B_i . When index i is absent from the word, we shall use the sharper estimate (2.5); this avoids a contribution proportional to $\mathbf{k}$.

Proof of Theorem 1.3. On the product probability space of B, U, W , let

$$F(B, U, W) = -\log q_{\Psi_W(B, U), \mathbf{m}}(W(\Psi_W(B, U))). \quad (3.2)$$

By Lemma 2.1, this has the law of $I_{\mathbf{m}}$. The underlying coordinates are $B_1, \dots, B_{\mathbf{k}}$, $U_1, \dots, U_{\mathbf{m}}$ and $W_1, \dots, W_{\mathbf{m}}$. Write $F^{B_i}, F^{U_j}, F^{W_j}$ for the values obtained by independently resampling the indicated coordinate. Every realised endpoint is reached by its sampled word, whose probability is $(2\mathbf{k})^{-\mathbf{m}}$. We therefore have the uniform bound

$$0 \leq F, F^{B_i}, F^{U_j}, F^{W_j} \leq \mathbf{m} \log(2\mathbf{k}). \quad (3.3)$$

¹⁹ B. Efron and C. Stein. "The jackknife estimate of variance". In: *Ann. Statist.* 9.3 (1981), pp. 586-596; J. M. Steele. "An Efron-Stein inequality for nonsymmetric statistics". In: *Ann. Statist.* 14.2 (1986), pp. 753-758. DOI: 10.1214/aos/1176349952.

Proposition 2.8 bounds the number of record values on each stated event by $M = (2km)^{Cb^2}$. Since $m \leq k$ and $b \leq C \log(2k)$, this gives

$$\log M \leq C \log^3(2k). \quad (3.4)$$

Together with the independence stated in Remark 2.2, this allows us to apply Lemma 2.6 to each of the three types of resampling.

Resampling U_j . A coordinate U_j with $j \notin S(W)$ has no effect on the tuple or on F . On the event that W is good and $j \in S(W)$, the record in Proposition 2.8 and Lemma 2.6 bound the contribution to $\mathbb{E}(F - F^{U_j})^2$ by $C(1 + \log^2 M) \leq C \log^6(2k)$. Only words that are not good remain; (3.3) bounds their contribution by $m^2 \log^2(2k) \mathbb{P}(W \notin \mathcal{G}_{k,m})$. Summing over $j = 1, \dots, m$ gives

$$\sum_{j=1}^m \mathbb{E}(F - F^{U_j})^2 \leq Cm \log^6(2k) + Cm^3 \log^2(2k) \mathbb{P}(W \notin \mathcal{G}_{k,m}). \quad (3.5)$$

Resampling W_j . When both words are good, the corresponding record gives the same bound $C \log^6(2k)$. Both words are uniform, so the event that at least one is not good has probability at most $2\mathbb{P}(W \notin \mathcal{G}_{k,m})$. Using (3.3) on this event and summing over the letters yields

$$\sum_{j=1}^m \mathbb{E}(F - F^{W_j})^2 \leq Cm \log^6(2k) + Cm^3 \log^2(2k) \mathbb{P}(W \notin \mathcal{G}_{k,m}). \quad (3.6)$$

Resampling B_i . It remains to bound the sum over i . If $N_i(W) = 1$, the construction does not use B_i and $F = F^{B_i}$. We treat absent and repeated indices separately.

Fix i . The event $N_i(W) = 0$ has probability $\alpha = (1 - 1/k)^m \geq 1 - m/k \geq 1/2$. For each z , let Q_z be the endpoint law of a uniform word conditioned on avoiding i . It is a mixture component of $q_{z,m}$ with weight α . On the event in question only $Z_i = B_i$ changes. Thus $Q_Z = Q_{Z'}$, and the sampled endpoint is unchanged. Comparing each unconditional probability with this common conditional probability and applying (2.5), we obtain

$$\mathbb{E}[(F - F^{B_i})^2 1_{\{N_i(W)=0\}}] \leq C(1 - \alpha) \leq Cm/k. \quad (3.7)$$

No restriction to good words is needed here. Summing over i gives at most Cm .

For the repeated indices, let $p_i = \mathbb{P}(W \in \mathcal{G}_{k,m}, N_i(W) \geq 2)$. We have $p_i \leq m/k$. By Proposition 2.8 and Lemma 2.6, the contribution on this event is at most $Cp_i(1 + \log^2(M/p_i))$, with value zero when $p_i = 0$.

To sum these bounds, use that $x \mapsto x \log^2(1/x)$ is increasing for $0 < x \leq m/k \leq 1/4096$. It follows that

$$\sum_{i=1}^k p_i (1 + \log^2(M/p_i)) \leq Cm(1 + \log^2 M + \log^2(k/m)) \leq Cm \log^6(2k).$$

There are at most m used indices in any word. For words that are not good, (3.7) already covers every unused index. By (3.3), the other terms contribute at most $Cm^3 \log^2(2k) \mathbb{P}(W \notin \mathcal{G}_{k,m})$ in total. Combining the three multiplicity cases gives

$$\sum_{i=1}^k \mathbb{E}(F - F^{B_i})^2 \leq Cm \log^6(2k) + Cm^3 \log^2(2k) \mathbb{P}(W \notin \mathcal{G}_{k,m}). \quad (3.8)$$

The three sums are now bounded by (3.5), (3.6) and (3.8).

Lemma 2.3 bounds their exceptional terms by

$Cm^{-37} \log^2(2k) \leq Cm \log^6(2k)$. Substituting these bounds in (3.1) for the representation (3.2) proves (1.9). $\square$

The proof requires bounds only on multiplicities and gaps between singleton positions. The expected number of pairs of positions with the same index may be large on the time scale considered below; the estimate still applies throughout the required range $m = o(k)$.

4. Cutoff

We now prove Theorem 1.1. We first obtain a spectral bound and locate the entropy threshold. The preliminary time bounds give cutoff when $\log k > (\log \log n)^2$; in the remaining range we use Theorem 1.3.

4.1 Collision and spectral estimates

For a probability measure q on G , write

$$\chi^2(q, \pi) = n \sum_{x \in G} q(x)^2 - 1. \quad (4.1)$$

Cauchy-Schwarz and $\log x \leq x - 1$ give

$$4\|q - \pi\|_{TV}^2 \leq \chi^2(q, \pi), \quad \log n - H(q) \leq \chi^2(q, \pi). \quad (4.2)$$

We first bound the probability that two independent words have the same value. We adapt Roichman's argument,²⁰ in the form used by

²⁰ Y. Roichman. "On random random walks". In: *Ann. Probab.* 24.2 (1996), pp. 1001-1011, Theorem 2.

Hermon and Olesker-Taylor.²¹ We give the estimate for the sampling convention in (1.1).

Lemma 4.1. Let G be a finite group of order n , and let k, m be positive integers with $1 \leq m \leq k$. Then

$$\mathbb{E}_Z \chi^2(q_{Z,m}, \pi) \leq (n-1) \binom{k}{m} \left(\frac{m}{k}\right)^{2m} \leq n \left(\frac{em}{k}\right)^m. \quad (4.3)$$

Proof. Let W, W' be independent uniform words of length m , independent of Z . Conditional on Z , their values are independent with law $q_{Z,m}$, so

$$\mathbb{E}_Z \chi^2(q_{Z,m}, \pi) = n \mathbb{P}(W(Z) = W'(Z)) - 1.$$

Let E be the event that no index occurs exactly once among the $2m$ letters. On E^c , choose such an index i and condition on the words and all generators other than Z_i . The equality $W(Z) = W'(Z)$ then has the form $gZ_i^\varepsilon h = e$ with g, h fixed. There is exactly one choice of Z_i for which this equality holds, so the conditional collision probability is $1/n$. Bounding the collision probability by one on E , we obtain $\mathbb{E}_Z \chi^2(q_{Z,m}, \pi) \leq (n-1) \mathbb{P}(E)$.

On E , at most m distinct indices occur. A union bound over all m -subsets of $[k]$ gives $\mathbb{P}(E) \leq \binom{k}{m} (m/k)^{2m}$, proving the first inequality. The second follows from $\binom{k}{m} \leq (ek/m)^m$. $\square$

The same collision estimate bounds the non-trivial eigenvalues of P_Z . Write $\|f\|_{2,\pi}$ for the $L^2(\pi)$ norm of a function f on G . The matrix P_Z is symmetric. Let $\lambda_*(z)$ be the largest absolute value of an eigenvalue of its restriction to the orthogonal complement of the constant functions. In particular, $\lambda_*(z) = 1$ if the Cayley graph is disconnected. The next lemma is a version of the Alon-Roichman spectral estimate²² for our sampling convention.

Lemma 4.2. Assume $n \rightarrow \infty$ and $k/\log n \rightarrow \infty$. Then $\mathbb{P}(\lambda_*(Z) > 1/2) \rightarrow 0$. On the event $\{\lambda_*(Z) \leq 1/2\}$, every mean-zero function f on G satisfies

$$\|e^{s(P_Z - I)} f\|_{2,\pi} \leq e^{-s/2} \|f\|_{2,\pi}, \quad s \geq 0. \quad (4.4)$$

In particular, the sampled generators generate G with probability tending to one.

²¹ J. Hermon and S. Olesker-Taylor. "Cutoff for almost all random walks on abelian groups". In: *J. Eur. Math. Soc.* 28 (2026), pp. 1913–1977, Section 7.2.

²² N. Alon and Y. Roichman. "Random Cayley graphs and expanders". In: *Random Structures Algorithms* 5.2 (1994), pp. 271–284, Theorem 1 and Section 2.1.

Proof. By symmetry and left-translation invariance,

$$\mathrm{tr}(P_z^{2m}) - 1 = nP_z^{2m}(e, e) - 1 = n \sum_x q_{z,m}(x)^2 - 1 = \chi^2(q_{z,m}, \pi).$$

It follows that $\lambda_*(z)^{2m} \leq \chi^2(q_{z,m}, \pi)$, since the even powers of the eigenvalues are non-negative. With $m = \lceil \log n \rceil$, we have $m \leq k$ for large n . Lemma 4.1 and Markov's inequality therefore give

$$\mathbb{P}(\lambda_*(Z) > 1/2) \leq n \left(\frac{4em}{k} \right)^m = o(1),$$

since $k/m \rightarrow \infty$. On the complementary event, all eigenvalues of $P_Z - I$ on the orthogonal complement of the constants are at most $-1/2$. Spectral decomposition gives (4.4). The Cayley graph is then connected, since otherwise $\lambda_*(Z) = 1$. $\square$

4.2 The entropy threshold

For the remainder of this section, assume (1.5) and write

$$L = \log n, \quad \rho = \log(k/L), \quad T_- = \frac{L}{\log k}, \quad T_+ = \frac{L}{\rho}. \quad (4.5)$$

Then $\rho \rightarrow \infty$, $\rho = o(L)$ and $T_-, T_+ \rightarrow \infty$. We obtain a lower bound for the mixing time by counting reachable vertices, and an upper bound from Lemma 4.1. The collision estimate also controls the expected entropy.

Proposition 4.3. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy $k/\log n \rightarrow \infty$ and $\log k/\log n \rightarrow 0$. For each G , let Z consist of k independent uniform elements of G . Then, for every fixed $\delta \in (0, 1)$,

$$\inf_{z \in G^k} d_z((1 - \delta)T_-) \rightarrow 1, \quad d_Z((1 + \delta)T_+) \xrightarrow{\mathbb{P}} 0. \quad (4.6)$$

Moreover, for every fixed $\eta > 0$,

$$L - h_{\lceil (1+\eta)T_+ \rceil} \rightarrow 0. \quad (4.7)$$

Proof. Set $t = (1 - \delta)T_-$ and $r = \lfloor (1 - \delta/2)T_- \rfloor$. There are at most $\sum_{j=0}^r (2k)^j \leq 2(2k)^r$ vertices reachable from e in at most r steps. A Poisson variable N_t of mean t satisfies $\mathbb{P}(N_t \leq r) \rightarrow 1$, whereas $2(2k)^r/n \rightarrow 0$. Thus, uniformly in z , the walk at time t lies with probability tending to one in a set whose uniform measure tends to zero. This proves the lower bound.

For the upper bound and the entropy estimate, fix $\eta > 0$ and let $m = \lceil (1+\eta)L/\rho \rceil$. Since $m = o(k)$, we may apply Lemma 4.1. To estimate its right-hand side, note that

$$\log \left[n \left(\frac{em}{k} \right)^m \right] = L - m \log \frac{k}{em} = -\eta L + o(L),$$

because $\log(k/(em)) = \rho + \log \rho - \log(1+\eta) - 1 + o(1)$. It follows from (4.2) that $\|q_{z,m} - \pi\|_{TV} \xrightarrow{\mathbb{P}} 0$ and $L - h_m \rightarrow 0$.

For each fixed z , the discrete-time distance from uniformity is non-increasing. Hence, for $t > m$,

$$d_z(t) \leq \mathbb{P}(N_t < m) + \|q_{z,m} - \pi\|_{TV}.$$

With $0 < \eta < \delta$ and $t = (1+\delta)T_+$, the Poisson tail tends to zero. This proves the upper bound in (4.6); the estimate $L - h_m \rightarrow 0$ gives (4.7). $\square$

For every finite group G , every $z \in G^k$ and every integer $m \geq 0$, we have

$$0 \leq H(q_{z,m+1}) - H(q_{z,m}) \leq \log(2k). \quad (4.8)$$

The lower bound follows from concavity, since the next-step distribution is a mixture of translates of $q_{z,m}$. The upper bound follows from subadditivity of entropy, since the next position is determined by the current position and the next letter. Hence h_m is non-decreasing and $0 \leq h_m \leq \min\{m \log(2k), \log n\}$.

Write

$$a_L = L^{2/3}. \quad (4.9)$$

We next locate t_* between the two time scales above.

Lemma 4.4. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ be a positive integer satisfying $k/\log n \rightarrow \infty$ and $\log k/\log n \rightarrow 0$. Then t_* in (1.4) is well defined for all sufficiently large n . For every fixed $\eta > 0$,

$$\frac{L - 2a_L}{\log(2k)} \leq t_* \leq \left\lceil (1+\eta) \frac{L}{\rho} \right\rceil \quad (4.10)$$

for all sufficiently large n . In particular, $t_* \rightarrow \infty$.

Proof. Proposition 4.3 gives $h_{\lceil (1+\eta)L/\rho \rceil} = L - o(1) > L - 2a_L$ for large n . Thus t_* exists and satisfies the upper bound. The inequality $h_m \leq m \log(2k)$ from (4.8) gives the lower bound, which tends to infinity since $\log k = o(L)$. $\square$

When $\log k > (\log L)^2$, the two time scales are asymptotically equivalent. The preceding bounds therefore give cutoff in this range.

Lemma 4.5. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy $k/\log n \rightarrow \infty$, $\log k/\log n \rightarrow 0$ and $\log k > (\log L)^2$. For each G , let Z consist of k independent uniform elements of G . Then

$$t_* \sim \frac{L}{\log k} \sim \frac{L}{\rho}, \quad (4.11)$$

and, for every fixed $\delta \in (0, 1)$,

$$d_Z((1 - \delta)t_*) \xrightarrow{\mathbb{P}} 1, \quad d_Z((1 + \delta)t_*) \xrightarrow{\mathbb{P}} 0.$$

Proof. Since $\log L/\log k \rightarrow 0$, we have $\rho/\log k = 1 - \log L/\log k \rightarrow 1$. For every fixed $\eta > 0$, Lemma 4.4 gives

$$(1 - o(1))\frac{L}{\log k} \leq t_* \leq (1 + \eta + o(1))\frac{L}{\rho}.$$

Letting $\eta \rightarrow 0$ proves (4.11). Since $T_- \sim T_+ \sim t_*$, Proposition 4.3 and monotonicity of the total variation distance give (1.6). $\square$

4.3 Concentration in continuous time

By Lemma 4.5, it remains to consider the range

$$\log k \leq (\log L)^2. \quad (4.12)$$

We impose this assumption in the next lemma and Proposition 4.7. For $t \geq 0$, let X_t have conditional law $p_{Z,t}$ given Z . We shall apply the following uniform estimate at the deterministic times $t_* - 1$ and t_* .

Lemma 4.6. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy $k/\log n \rightarrow \infty$, $\log k/\log n \rightarrow 0$ and $\log k \leq (\log L)^2$. For each G , let Z consist of k independent uniform elements of G , and let X_t have conditional law $p_{Z,t}$ given Z . Then, uniformly for deterministic times

$$\frac{L}{2 \log(2k)} \leq t \leq \frac{5L}{2\rho}, \quad (4.13)$$

we have

$$-\log p_{Z,t}(X_t) = h_{[t]} + o_{\mathbb{P}}(a_L) \quad (4.14)$$

under the joint law of Z and X_t .

Here uniformity means that, for every fixed $\varepsilon > 0$, the supremum over (4.13) of the probability that the absolute error exceeds εa_L tends to zero.

Proof. Let t satisfy (4.13). All estimates below are uniform in t . Let N be Poisson with mean t , independent of Z , and, conditional on $N=m$ and Z , let X have law $q_{Z,m}$. Then X has conditional law $p_{Z,t}$. By Chebyshev's inequality,

$$\mathbb{P}(|N-t| > \sqrt{t} \log L) \leq (\log L)^{-2}. \quad (4.15)$$

On the complementary event, N belongs to the interval

$$\frac{L}{4 \log(2k)} \leq m \leq \frac{3L}{\rho} \quad (4.16)$$

for all sufficiently large n . Indeed, the lower bound on t is at least a constant multiple of $L/(\log L)^2$, so $\sqrt{t} \log L = o(t)$ uniformly. For integers m in (4.16), we have $m/k \rightarrow 0$ and $m \rightarrow \infty$ uniformly. Since $\mathbb{E}l_m = h_m$, Theorem 1.3 gives

$$\mathbb{E}(l_m - h_m)^2 \leq CL(\log L)^{12} = o(a_L^2). \quad (4.17)$$

Together with (4.15), this implies $-\log q_{Z,N}(X) = h_N + o_{\mathbb{P}}(a_L)$. Moreover, on the same event, (4.8) gives

$$|h_N - h_{\lfloor t \rfloor}| \leq (|N-t|+1) \log(2k) \leq C\sqrt{L}(\log L)^3 = o(a_L).$$

It follows that $-\log q_{Z,N}(X) = h_{\lfloor t \rfloor} + o_{\mathbb{P}}(a_L)$.

It remains to compare $q_{Z,N}(X)$ with $p_{Z,t}(X)$. Write $\alpha_m = e^{-t} t^m / m!$. When $|N-t| \leq \sqrt{t} \log L$, we have

$$-\log \alpha_N \leq C(\log L)^2. \quad (4.18)$$

Indeed, $\log(m!) = m \log m - m + O(\log(m+1))$ gives $-\log \alpha_m = t - m + m \log(m/t) + O(\log(m+1))$. When $m/t \rightarrow 1$, the first three terms are at most $C(m-t)^2/t$, proving (4.18) uniformly on this event.

Since $p_{Z,t} \geq \alpha_m q_{Z,m}$, Lemma 2.4 bounds the second moment of $\log(p_{Z,t}(X)/q_{Z,m}(X))$, conditional on $N=m$ and $Z=z$, by $\log^2(1/\alpha_m) + 2$. Equations (4.15) and (4.18) therefore give $\log(p_{Z,t}(X)/q_{Z,N}(X)) = o_{\mathbb{P}}(a_L)$. Combining this with the estimate for $-\log q_{Z,N}(X)$ proves (4.14). $\square$

4.4 The cutoff window

We now combine Lemma 4.6 with the spectral estimate to bound the total variation distance on either side of the entropy threshold. We adapt Salez's entropy bounds²³ to the deterministic centre t_* .

²³ J. Salez. "Cutoff for non-negatively curved Markov chains". In: *J. Eur. Math. Soc.* 26.11 (2024), pp. 4375–4392, Lemmas 7–8.

Proposition 4.7. Let G range over a sequence of finite groups with $n = |G| \rightarrow \infty$, and let $k = k(G)$ satisfy $k/\log n \rightarrow \infty$, $\log k/\log n \rightarrow 0$ and $\log k \leq (\log L)^2$. For each G , let Z consist of k independent uniform elements of G . Then

$$\mathbb{E}_Z d_Z(t_* - 1) \rightarrow 1, \quad \mathbb{E}_Z d_Z(t_* + 4a_L) \rightarrow 0. \quad (4.19)$$

Moreover, $a_L = o(t_*)$.

Proof. Lemma 4.4, with $\eta = 1$, places $t_* - 1$ and t_* in (4.13) for large n . It also gives

$$\frac{a_L}{t_*} \leq \frac{L^{2/3} \log(2k)}{L - 2L^{2/3}} \leq C \frac{(\log L)^2}{L^{1/3}} \rightarrow 0. \quad (4.20)$$

For the lower bound, set $t = t_* - 1$ and let $E_Z = \{x : p_{z,t}(x) > e^{a_L}/n\}$. Minimality of t_* gives $h_t < L - 2a_L$, so Lemma 4.6 yields $\mathbb{P}(X_t \in E_Z) \rightarrow 1$. On the other hand, $\sum_x p_{z,t}(x) = 1$ gives $\pi(E_Z) \leq e^{-a_L}$. Thus

$$\mathbb{E}_Z d_Z(t) \geq \mathbb{P}(X_t \in E_Z) - e^{-a_L} \rightarrow 1.$$

For the upper bound, $h_{t_*} \geq L - 2a_L$, so Lemma 4.6 gives $\mathbb{P}(p_{Z,t_*}(X_{t_*}) > e^{3a_L}/n) \rightarrow 0$. For each z , define

$$v_z(x) = p_{z,t_*}(x) 1_{\{p_{z,t_*}(x) \leq e^{3a_L}/n\}}, \quad \beta_z = \sum_x v_z(x). \quad (4.21)$$

Then $\mathbb{E}_Z(1 - \beta_Z) \rightarrow 0$. The discarded part contributes at most $1 - \beta_z$ to the distance from uniformity. It remains to bound the distance of the retained measure from $\beta_z \pi$ after time $4a_L$. Its density $f_z = n v_z$ with respect to π is bounded by e^{3a_L} , so

$$\|f_z - \beta_z\|_{2,\pi}^2 = \int f_z^2 d\pi - \beta_z^2 \leq e^{3a_L} \beta_z \leq e^{3a_L}. \quad (4.22)$$

On the event $\{\lambda_*(z) \leq 1/2\}$, symmetry and Lemma 4.2 give

$$\|v_z e^{s(P_z - I)} - \beta_z \pi\|_{TV} \leq \frac{1}{2} e^{-s/2} \|f_z - \beta_z\|_{2,\pi} \leq \frac{1}{2} e^{-s/2 + 3a_L/2}.$$

Taking $s = 4a_L$, we obtain $d_z(t_* + 4a_L) \leq 1 - \beta_z + \frac{1}{2} e^{-a_L/2}$ on this event. Its complement has probability $o(1)$ by Lemma 4.2. Since $d_z \leq 1$ and $\mathbb{E}_Z(1 - \beta_Z) \rightarrow 0$, averaging proves the upper bound. $\square$

Proof of Theorem 1.1. Lemma 4.4 shows that t_* is well defined and tends to infinity. Lemma 4.5 proves the required convergence along the part of the sequence for which $\log k > (\log L)^2$.

On the remaining part, Proposition 4.7 gives (4.19) and $a_L = o(t_*)$. Fix $\delta \in (0, 1)$. For all sufficiently large indices in this range, $(1 - \delta)t_* \leq t_* - 1$ and $(1 + \delta)t_* \geq t_* + 4a_L$. By monotonicity,

$$\mathbb{E}_Z[1 - d_Z((1 - \delta)t_*)] \rightarrow 0, \quad \mathbb{E}_Z d_Z((1 + \delta)t_*) \rightarrow 0.$$

Markov's inequality gives both convergences in probability on this part of the sequence. The two ranges exhaust the sequence, completing the proof. $\square$